

Micro Energy Harvesting 2026

Exercise 3: Thermoelectric converters

Task 1: TEGs - Thermoelectric Generators

a) How many thermocouples are needed to obtain a desired output voltage of 2.5 V, for example to operate a microprocessor, from a temperature difference of $\Delta T = 10$ K by 20 °C? Determine the required number of unicouples for the following material combinations:

- Nichrome (NiCr alloy with additions of Mn, Fe, Si, and C) & Constantan (CuNi alloy)
- n- Bi_2Te_3 & p- $(\text{Bi,Sb})_2\text{Te}_3$

$S_{\text{Nichrome}} = 25 \mu\text{V/K}$ (+5 $\mu\text{V/K}$ for Pt); $S_{\text{Constantan}} = -35 \mu\text{V/K}$ (+5 $\mu\text{V/K}$ for Pt) $\Rightarrow S_{\text{Nichrome-Constantan}} = 60 \mu\text{V/K}$ (Wikipedia)

$$U = m \cdot (S_p - S_n) \cdot \Delta T \Rightarrow m = U / (S_{pn} \cdot \Delta T) = 2.5 \text{ V} / (60 \mu\text{V/K} \cdot 10 \text{ K})$$

$$@ 10 \text{ K } \Delta T \Rightarrow 600 \mu\text{V} = 0,6 \text{ mV} \quad \Rightarrow m_{\text{Nichrome-Constantan}} = 4167$$

$S_{\text{BiTe}} = -240 \mu\text{V/K}$; $S_{\text{SbBiTe}} = 225 \mu\text{V/K} \Rightarrow S_{\text{BiTe,SbBiTe}} = 465 \mu\text{V/K}$ (Handbook of Thermoelectrics 2006)

$$@ 10 \text{ K } \Delta T \Rightarrow 4650 \mu\text{V} = 4,65 \text{ mV} \quad \Rightarrow m_{\text{BiTe,SbBiTe}} = 538$$

b) A typical TEG, such as the Thermalforce TEG 049-150-30 has a 25x25 mm² footprint and consists of 49 thermocouples of Bi_2Te_3 & $(\text{Bi,Sb})_2\text{Te}_3$. How large would the footprint and edge length of a similar generator with 257 thermocouples be?

rough calculation incl. TE material and empty space

$$A_{\text{Thermocouple}} = 25 \times 25 \text{ mm}^2 = 625 \text{ mm}^2 \quad \Rightarrow 625 \text{ mm}^2 / 49 = 12,755 \text{ mm}^2 \text{ per TC}$$

$$\Rightarrow A_{257} = 3278 \text{ mm}^2 \quad \Rightarrow \sqrt{A_{257}} \quad \Rightarrow L_{257} = 57,3 \text{ mm} \sim 58 \text{ mm}$$

c) Using the data sheet, you can identify the values for the thermal resistance (or conductivity) of the generator at 300 K, as well as the Seebeck voltage $\alpha_{\text{generator}}$. Use these values to calculate ZT @ 293 K.

We do NOT use corrected values, because we do not calculate between 150 °C and 30 °C but at 300 K!

$$R_{\text{Gen}} = 0,57 \text{ Ohm}$$

$$\alpha_{\text{Gen}} = 0,016 \text{ V/K}$$

$$R_{\text{th,Gen}} = 5,263 \text{ K/W}$$

$$ZT_{\text{Gen}} = (\alpha_{\text{Gen}})^2 \cdot R_{\text{th,Gen}} / R_{\text{Gen}} \cdot T = 0,002364 \text{ 1/K} \cdot 293 \text{ K} = 0,69$$

d) A typical thermal conductivity value for Bi_2Te_3 is $\kappa = 2 \text{ W/(K}\cdot\text{m)}$. Use this value to determine the ratio of leg-length to footprint $L_{\text{Leg}}/A_{\text{Leg}}$ of a single thermo leg. Using this value, you can additionally calculate the ratio of active area A_{act} to total area A_{Gen} of this TEG. The length of the legs can be calculated from the data sheet.

$$A_{\text{total}} = 625 \text{ mm}^2$$

$$L_{\text{Leg}} = 1.7 \text{ mm} \text{ (3.7 mm - 2 * 1 mm Al}_2\text{O}_3 \text{ from data sheet)}$$

Based on Generator thermal resistance $R_{\text{th,Gen}} = 5,263 \text{ K/W}$ and thermal conductivity κ_{BiTe}

$$R_{\text{th,Gen}} = L_{\text{Leg}} / (\kappa_{\text{BiTe}} \cdot A_{\text{act}}) \quad \text{mit } A_{\text{akt}} = 2 \cdot 49 \cdot A_{\text{Leg}}$$

$$\Rightarrow R_{\text{th,Gen}} = L_{\text{Leg}} / (\kappa_{\text{BiTe}} \cdot A_{\text{Leg}} \cdot 2 \cdot 49) \quad \Rightarrow A_{\text{Leg}} = L_{\text{Leg}} / (R_{\text{th,Gen}} \cdot 2 \cdot 49 \cdot \kappa_{\text{BiTe}}) = 1,648 \text{ mm}^2$$

$$\Rightarrow L_{\text{Leg}} / A_{\text{Leg}} = 1,03 \text{ 1/mm} \text{ (appr. 1)}$$

$$A_{act}/A_{total} = 2 \cdot 49 \cdot A_{Leg} / A_{total} = \underline{0,258} (25,8 \%)$$

Task 2: μ TEGs

Using the data sheet for the Micropelt "MPG-D751" μ -thermogenerator, calculate ZT @ 293 K and L/A ratios for the thermo legs and the ratio of active area to generator area as before. The specific electrical resistance can be assumed with the typical value of $\rho = 10^{-5} \Omega\text{m}$. The area of the thermo legs can be taken from the data sheet. Are these values significantly different from the macro generator? How high (long) are the thermolegs and why are they not made longer?

Extraction of the values:

Only the thermoelectrically active part is interesting - without contact surfaces [alternative including contact surfaces]. All dimensions averaged, since we are looking for the average generator (arguments for min. (worst case) or max. (ideal case) could be made as well)

$$Top_{av}: 3,388 * 3,339 \text{ mm}^2$$

$$Bottom_{av}: 4,223 * 3,339 \text{ mm}^2$$

$$R_{el,Gen} = 300 \Omega$$

$$A_{total} = (3,388 * 3,339) \text{ mm}^2 = 11,3125 \text{ mm}^2 [14,1006 \text{ mm}^2]$$

$$A_{Leg} = (35 \mu\text{m})^2 = 1225 \mu\text{m}^2 = 0,001225 \text{ mm}^2 = 1,225 * 10^{-3} \text{ mm}^2$$

$$R_{el,Gen} = \rho_{BiTe} * L_{total} / A_{Leg} = \rho_{BiTe} * L_{Leg} * 2 * 540 / A_{Leg}$$

$$\Rightarrow L_{Leg} / A_{Leg} = R_{el,TEG} / (2 * 540 * \rho_{BiTe}) = 300 \Omega / 0,0108 \Omega\text{mm} = 27,7778 \text{ 1/mm} = 0,027778 \text{ 1/}\mu\text{m}$$

$$\Rightarrow L_{Leg} = L_{Leg} / A_{Leg} * A_{Leg} = 1225 \mu\text{m} * 0,027778 \text{ 1/}\mu\text{m} = 34 \mu\text{m}$$

$$\text{Standard TEG: } L_{Leg} \approx 1\text{-}2 \text{ mm; } L_{Leg} / A_{Leg} \approx 1 \Rightarrow \text{BIG difference!!!}$$

$$A_{act} = 2 * 540 * A_{Leg} = 1080 * 0,001225 \text{ mm}^2 = 1,323 \text{ mm}^2$$

$$A_{act} / A_{gen} = 1,323 \text{ mm}^2 / 11,3125 \text{ mm}^2 = 0,117 [0,094] (11.7 \% - \text{smaller, but similar})$$

$$ZT = (0,140 \text{ V/K})^2 * 12,5 \text{ K/W} * 293 \text{ K} / 300 \Omega = 0,239 [0,245 @ 300 \text{ K}] (\text{significantly smaller})$$

Length = 34 μm (or 40 μm according to the height specifications in the data sheet), process-related no extension possible because of the pyramid shape.

Task 3: Thermoelectric Power Generation

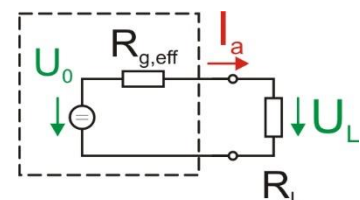
A typical DC-DC converter such as the one from Linear Technologies (bipolar, LTC3109, $U_{start} = 27\text{mV}$) is the only load connected to a TEG and has an internal resistance of 3Ω . Calculate the output power (at the load resistance) for a generator with 144 thermocouples (Seebeck coefficient: $400 \mu\text{V/K}$, internal resistance $R_g = 10 \Omega$, thermal resistance $K_g = 6.1 \text{ K/W}$), which has the heat sink of **Task 4** ($K_{HS} = 4 \text{ K/W}$) coupled to its hot side to generate a temperature difference between heat sink and cold side of the generator of 15 K . Also calculate the output voltage at the generator. At which temperature difference between the heat sink and the cold side does the DC-DC converter start to run?

Without DC-DC efficiency, since the calculation is done for the load resistance

$$P_{el} = \left(\frac{R_{th,TEG}}{R_{th,TEG} + R_{HS}} \right)^2 \cdot \frac{(m \cdot S \cdot \Delta T_{System})^2}{R_L} \cdot \left(\frac{R_L}{R_{g,eff} + R_L} \right)^2$$

$$\Rightarrow P_{el} = (6,1 \text{ K/W} / (6,1 \text{ K/W} + 4 \text{ K/W}))^2 * (144 * 0,0004 \text{ V/K} * 15 \text{ K})^2 / 3 \Omega * (3 \Omega / 10 \Omega + 3 \Omega)^2 = 0,3648 * 0,24883 \text{ W} * 0,05325 = 0,0048 \text{ W} = 4,8 \text{ mW}$$

$$P_{el} = (U_L)^2 / R_L \Rightarrow U_L = (P_{el} * R_L)^{-1} = (4,8 \text{ mW} / 3 \Omega)^{-1} = 0,12 \text{ V}$$



Alternative solution: TEG as voltage divider:

$$\Delta T_{TEG} = \left(\frac{R_{th,TEG}}{R_{th,TEG} + R_{HS}} \right) \cdot \Delta T \quad \Rightarrow \Delta T_{TEG} = 0,60396 \cdot 15 \text{ K} = 9,0594 \text{ K}$$

$$\Rightarrow U_{0,real} = m \cdot S \cdot \Delta T_{TEG} = 144 \cdot 0,0004 \text{ V/K} \cdot 9,0594 \text{ K} = 0,52 \text{ V}$$

$$\frac{U_L}{U_{0,real}} = \frac{R_L}{R_{g,eff} + R_L} \quad \Rightarrow U_L = U_{0,real} \cdot \left(\frac{R_L}{R_{g,eff} + R_L} \right) = 0,52 \text{ V} \cdot (3 \Omega / 10 \Omega + 3 \Omega) = \underline{0,12 \text{ V}}$$

$$\Delta T_{min} = 0,027 \text{ V} \cdot 15 \text{ K} / 0,12 \text{ V} = \underline{3,38 \text{ K}} \text{ (from } 0,12 \text{ V @ } 15 \text{ K)}$$

Task 4: TEG Optimization

A heat sink from Fischer electronic (ICK S 25 x 25 x 18) is to be used in a busy road tunnel together with an **ideal TEG**. By constantly passing cars and trucks, forced convection occurs on average, so that the heat sink has a thermal resistance of 4 K/W.

a) The generator is a standard macroscopic TEG, with a leg length of $L=1.7 \text{ mm}$ and the leg area $A=1.65 \text{ mm}^2$. How many thermocouples are needed so that the thermogenerator has the same thermal resistance as the heat sink (neglect contact resistances, material Bi_2Te_3)?

$$R_{th,Gen} = 4 \text{ K/W}$$

$$\kappa_{\text{BiTe}} = 2 \text{ W/(K m)} \text{ from Task 1}$$

$$R_{th,Leg} = L / (\kappa_{\text{BiTe}} \cdot A) = 1.7 \text{ mm} / (2 \text{ W/m K} \cdot 1.65 \text{ mm}^2) = 515 \text{ K/W}$$

$$n_{Leg} = R_{th,Leg} / R_{th,Gen} = 515 / 4 = 128,75 \quad \Rightarrow 128,75 / 2 = 64,4 - \underline{\text{about 65 Thermocouples}}$$

b) What is the ratio of active to total area?

$$A_{total} \text{ based on heat sink dimensions: } A_{Gen} = 25 \text{ mm} \times 25 \text{ mm} = 625 \text{ mm}^2$$

$$A_{act} = 2 \cdot 65 \cdot A_{Leg} = 214,5 \text{ mm}^2$$

$$A_{act}/A_{total} = 214,5 \text{ mm}^2 / 625 \text{ mm}^2 = \underline{0,343}$$

c) Adjust the area A of the legs, so that 144 thermocouples fit on the generator (25 % active area), how long would the thermo legs have to be to adjust the generator thermally ideal and what is their edge length?

$$R_{th,Gen} = 4 \text{ K/W}$$

$$A_{total} = 625 \text{ mm}^2$$

$$A_{act} = 2 \cdot 144 \cdot A_{Leg} \quad \Rightarrow A_{Leg} = 0,25 \cdot A_{total} / (2 \cdot 144) = 156,25 \text{ mm}^2 / 288 = \underline{0,54 \text{ mm}^2}$$

$$\text{Edge length: } \sqrt{A_{Leg}} = \sqrt{0,54 \text{ mm}^2} = \underline{0.735 \text{ mm}}$$

$$R_{th,Gen} = L_{Leg} / (\kappa_{\text{BiTe}} \cdot A_{act}) \quad \text{with } A_{act} = 2 \cdot 144 \cdot A_{Leg} \quad \Rightarrow R_{th,Gen} = L_{Leg} / (\kappa_{\text{BiTe}} \cdot A_{Leg} \cdot 2 \cdot 144)$$

$$\Rightarrow L_{Leg} = R_{th,Gen} \cdot \kappa_{\text{BiTe}} \cdot A_{Leg} \cdot 2 \cdot 144 = 4 \text{ K/W} \cdot 2 \text{ W/(K m)} \cdot 0,54 \text{ mm}^2 \cdot 288 = \underline{1,24 \text{ mm}}$$

d) What is the internal resistance of this generator?

$$\rho_{\text{BiTe}} = 10^{-5} \Omega \text{m (from Task 2)}$$

$$R_{el,Gen} = \rho_{BiTe} * L_{total} / A_{Leg} = \rho_{BiTe} * 2 * 144 * L_{Leg} / A_{Leg} = 10^{-5} \Omega m * 288 * 1,24 \text{ mm} / 0,54 \text{ mm}^2 = \underline{6,61 \Omega}$$

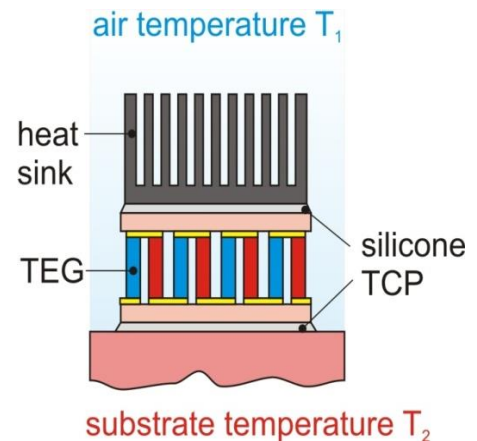
For comparison:

$$R_{el,Gen} (65 \text{ TC}) = 1,34 \Omega$$

$$R_{el,Gen} (49 \text{ TC, Thermalforce}) = 0,54 \Omega$$

Task 5: Thermoelectric Harvesting System

The following model of a wall-mounted system is given. The Fischer Elektronik ICK S 25 x 25 x 18 heat sink (Task 4) is used together with Micropelt D651 micro generators to harvest from the temperature difference between warm air and cold wall ($T_{cold} = 27.5 \text{ }^{\circ}\text{C}$). As the heat sink is much larger than the individual generator, it is decided to equip the heat sink with as many generators of this type as possible. The air speed of 9 km/h on this hot day is due to a radial fan set up, which blows the warm air ($T_{hot} = 35.5 \text{ }^{\circ}\text{C}$) through the fins. The contact resistances due to the thermal paste between heat sink and generator or generator and wall are $K_{TCP} = 0.05 \text{ K/W}$.



Calculate the output voltage of the generators without load when operated in parallel? How high is it when used serially? Also calculate the output power for both cases for a load resistance of $R_L = 100 \Omega$. Any comments on the practicality of this system?

Features of the D651 (from data sheet):

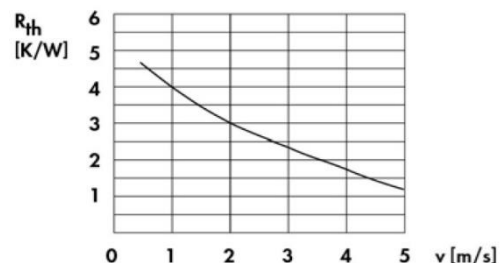
$$K_{D651} = 22 \text{ K/W}$$

$$U_{D651} = 75 \text{ mV/K}$$

$$R_{D651} = 185 \Omega$$

Thermal resistance KHS of the heat sink according to the graph given in the script or data sheet:

$$K_{HS}(v=2,5 \text{ m/s}) = \underline{2,75 \text{ K/W}}$$



Max. Number of generators possible:

$$A_{D651} = 3,375 \text{ mm} * 2,5 \text{ mm} = \underline{8,44 \text{ mm}^2}$$

$$A_{HS} = 25 \text{ mm} * 25 \text{ mm} = \underline{625 \text{ mm}^2}$$

$$n_{max} = 625 / 8,44 = 74,05 \approx \underline{74}$$

Alternative (better) solution via the individual edges of the μ TEGs:

$$n_{max,A} = L_{HS} / L_{D651,A} = 25 / 3,375 = 7,4 \Rightarrow$$

$$n_{max,B} = L_{HS} / L_{D651,B} = 25 / 2,5 = 10$$

$$\underline{n_{max} = 70}$$

Thermal resistance of all TEGs:

$$K_{total} = K_{D651} / 74 = \underline{0,297 \text{ K/W}} [0,314 \text{ K/W for 70 TEGs}]$$

Thermal Conductive Paste

$$K_{TCP}=0,05 \text{ K/W}$$

Temperature drop via TEG:

$$\Delta T = 8 \text{ K (external)}$$

$$\Delta T_{Gen} = \Delta T * K_{total} / (K_{total} + 2 * K_{TCP} + K_{HS}) = 8 \text{ K} * 0,094 [0,099] = \underline{0,755 \text{ K}} [0,794 \text{ K}]$$

Voltage in parallel corresponds to voltage of a single generator under the given ΔT :

$$U_{0,par} = 0,755 \text{ K} * 75 \text{ mV / K} = \underline{56,6 \text{ mV}} [59,5 \text{ mV}]$$

Voltage in series adds the voltages of all single generator together:

$$U_{0,ser} = U_{0,par} * n = \underline{4,188 \text{ V}} [4,165 \text{ V}]$$

Output voltage at load resistor $R_L = 100 \Omega$ (voltage divider):

$$U_{0,par,L} = U_{0,par} * (R_L / ((R_{D651}/74) + R_L)) = U_{0,par} * (100 / ((185/74) + 100)) = 56,6 \text{ mV} * 0,9756 = \underline{55,22 \text{ mV}} [0,9743; 57,97 \text{ mV}]$$

$$P_{par} = (U_{0,par,L})^2 / R_L = \underline{30,49 \mu W} [33,60 \mu W]$$

$$U_{0,ser,L} = U_{0,ser} * (R_L / (74 * R_{D651} + R_L)) = U_{0,ser} * (100 / (74 * 185 + 100)) = 4,188 \text{ V} * 0,00725 = \underline{30,36 \text{ mV}} [0,00766; 31,90 \text{ mV}]$$

$$P_{ser} = (U_{0,ser,L})^2 / R_L = \underline{9,22 \mu W} [10,18 \mu W]$$

Practicality Issues

- Connections between all the μ TEGs are difficult
- Proper alignment and flat fixation of all μ TEGs could be difficult
- **Price per μ TEG:** 100-200 € per piece => 7400 € [7000 €] – 14.800 € [14.000 €] – **a small car!!!**